Standard Operating Procedure

Prevention of Central Line-Associated Bloodstream Infections (CLABSI)

Illustrative / synthetic SOP created for training and demonstration purposes (e.g., testing retrieval-augmented generation systems). Does not represent the actual policy of any institution and must not be used to guide real clinical practice.

1. Purpose

This Standard Operating Procedure (SOP) defines the required steps for insertion, maintenance, and monitoring of central venous catheters (CVCs) to minimize the risk of Central Line-Associated Bloodstream Infection (CLABSI) across inpatient units.

2. Scope

Applies to all clinical staff involved in the insertion, care, and removal of central venous catheters, including internal jugular, subclavian, femoral, and peripherally inserted central catheter (PICC) lines, in adult inpatient units and intensive care units.

3. Roles and Responsibilities

Role	Responsibility	Frequency / Trigger
Inserting physician	Performs insertion using full barrier precautions; completes insertion checklist	At time of insertion
Bedside nurse	Performs daily site assessment, dressing checks, and documents line necessity	Every shift
Charge nurse / unit lead	Reviews line necessity during daily rounds; escalates lines with no active indication	Daily
Infection prevention team	Conducts surveillance, audits compliance, reports CLABSI rate to unit leadership	Monthly

4. Insertion Procedure

The following steps must be completed and documented on the central line insertion checklist for every catheter placement:

1. Confirm indication for central line placement and consider alternatives (e.g., peripheral IV, midline catheter) where clinically appropriate.
2. Perform hand hygiene using alcohol-based hand rub or soap and water for a minimum of 20 seconds.
3. Select insertion site based on lowest infection and mechanical complication risk; subclavian site is generally preferred over femoral for non-dialysis catheters when feasible.
4. Prepare skin with an alcoholic chlorhexidine (>0.5%) solution and allow it to fully dry before puncture.
5. Use maximal sterile barrier precautions: sterile gown, sterile gloves, mask, cap, and a full-body sterile drape.
6. Use ultrasound guidance for internal jugular and, where available, subclavian access to reduce insertion attempts.
7. Secure the catheter and apply a sterile, transparent semi-permeable dressing (or gauze if there is bleeding/oozing at the site).

8. Document date, time, site, operator, number of attempts, and any complications in the medical record and on the line insertion checklist.

5. Daily Maintenance Bundle

The following elements must be assessed and documented at least once per nursing shift:

- Daily review of continued line necessity — remove promptly when no longer clinically indicated
- Visual inspection of the insertion site for erythema, tenderness, swelling, or purulent drainage
- Dressing integrity — transparent dressings changed every 7 days or immediately if soiled, loose, or damp; gauze dressings changed every 2 days
- Hub and connector disinfection ("scrub the hub" for a minimum of 15 seconds) before every access
- Confirmation that administration sets are changed per protocol (typically every 96 hours for standard fluids, every 24 hours for lipid-containing or blood product infusions)

6. Line Removal Criteria

A central line should be removed as soon as any of the following applies:

- No active clinical indication for the line during daily multidisciplinary rounds
- Signs of local site infection (erythema, purulence) or suspected catheter-related bloodstream infection
- Catheter malfunction, dislodgement, or mechanical complication
- Patient transitioned to an alternative, lower-risk access device

7. Surveillance and Reporting

Each unit must report central line utilization ratio (central line days / patient days) and CLABSI rate per 1,000 central line days on a monthly basis to the infection prevention committee. Any confirmed CLABSI event should trigger a case review within 5 business days to identify contributing factors and corrective actions.

8. Escalation Pathway

9. Bedside nurse identifies a suspected site infection or unexplained fever in a patient with a central line.
10. Bedside nurse notifies the responsible physician and documents findings in the nursing notes.
11. Physician assesses the patient, orders blood cultures (peripheral and, if applicable, through the catheter), and determines whether line removal is indicated.
12. If CLABSI is confirmed or strongly suspected, infection prevention team is notified within 24 hours for surveillance logging.
13. Unit conducts a brief case review to identify any deviation from the maintenance bundle.

9. Training and Competency

All staff inserting or maintaining central lines must complete an annual competency assessment covering aseptic technique, the maintenance bundle, and this SOP. New staff must complete competency validation before independently performing central line insertion or maintenance.

10. Special Considerations by Catheter Type

Catheter Type	Typical Dwell Time	Special Notes
Non-tunneled CVC	Short-term (days)	Highest infection risk category; remove as soon as clinically feasible
PICC line	Weeks to months	Requires weekly dressing change if

		gauze, up to 7 days if transparent; measure arm circumference if suspected thrombosis
Tunneled cuffed catheter	Months	Cuff promotes tissue in-growth, reducing migration and infection risk over long-term use
Implanted port	Months to years	Access only with non-coring (Huber) needle; flush per protocol when not in active use

11. Documentation Requirements

The following must be recorded in the patient's chart for every central line episode:

- Date and time of insertion, and the indication for line placement
- Insertion site, operator name, number of attempts, and any immediate complications
- Daily necessity assessment and site condition, documented each shift
- Date of each dressing change and administration set change
- Date, time, and reason for line removal, including condition of the site at removal

12. Quality Improvement and Feedback

Unit-level CLABSI rates and bundle compliance audit results should be shared with frontline staff on a monthly basis through unit huddles or posted scorecards. Units with rising CLABSI rates or declining bundle compliance should conduct a focused root-cause analysis and implement a corrective action plan in partnership with the infection prevention team.

Appendix A: Quick-Reference Checklist

- Hand hygiene performed before procedure
- Maximal sterile barrier precautions used
- Chlorhexidine skin prep, allowed to dry
- Optimal site selected (subclavian preferred where feasible)
- Insertion checklist completed and filed
- Daily necessity review documented
- Dressing and hub care per schedule

Appendix B: Frequently Encountered Deviations

Deviation	Risk	Corrective Action
Dressing not changed on schedule	Increased colonization and infection risk	Nurse educator reinforcement; add reminder to shift checklist
Hub not scrubbed before access	Introduction of organisms into bloodstream	Direct observation audits; just-in-time coaching
Line left in without documented necessity	Prolonged unnecessary exposure to infection risk	Daily rounding checklist with explicit necessity field
Non-sterile technique noted during insertion	Elevated CLABSI risk from time of insertion	Immediate case review; re-training before next unsupervised insertion

Appendix C: Related Policies

- Hand Hygiene Policy (Institutional Policy IC-001)
- Peripheral IV and Midline Catheter Care SOP (Institutional Policy IC-004)
- Isolation Precautions Policy (Institutional Policy IC-010)
- Blood Culture Collection Procedure (Institutional Policy LAB-014)

End of document. Document ID: MED-SOP-003 (synthetic training corpus).